

Diana Zhu

858-340-0822 | dianazhu9879@gmail.com | linkedin.com/in/diana-zhu

EDUCATION

University of California San Diego

Bachelor of Science in Mathematics - Computer Science - GPA 4.0

La Jolla, CA

Sept. 2025 – May 2028

University of Pennsylvania

Pre-College Program - GPA 4.0

Philadelphia, PA

June 2024 – August 2024

École Dr. Charles Best Secondary School

Francophone Dual Dogwood Diploma - GPA 4.0

Coquitlam, BC

Sept. 2021 – May 2025

EXPERIENCE

Internship - AI Engineer

ArcellAI

January 2026 – Present

San Francisco, California

- Building a Perturbation Simulator using scGPT foundation models and retrieval methods to predict post-treatment cell states across a knowledge base of 33M+ cells
- Engineered a neural network adapter to predict embedding shifts from CRISPR and drug perturbations using PyTDC datasets
- Developed an automated pipeline for h5ad genomic data to enable high-throughput analysis and experimental reasoning

Researcher

Triton Quantitative Trading, UC San Diego

October 2025 – Present

La Jolla, California

- Developed a quantitative sector momentum strategy that achieved an 18.18% CAGR by rotating capital into top-performing S&P 500 sectors
- Applied OLS regression and to decompose returns, proving outperformance was by tactical sector bets
- Engineered a backtesting pipeline in QuantConnect to validate momentum persistence across market regimes

Internship - Team Leader

Shanghai Benyi Capital Investment Management Co., Ltd

June 2024 – August 2024

Pudong, Shanghai

- Collaborated on adapting and implementing optimized machine learning pipelines in Python using PyTorch, scikit-learn, NumPy, and pandas to classify 30K+ financial text entries.
- Fine-tuned and compared Random Forest, TextCNN, and LSTM models, improving F1-score by 5% and reaching 97% accuracy (98.9% with BERT)
- Delivered the model to company executives; accepted by the CEO; currently exploring internal applications

PROJECTS

Severless Financial Terminal | Python, React, AWS (S3, Lambda, CloudFront), CSS

January 2026 – Present

- Building a full-stack platform using AWS (S3, Lambda, DynamoDB) to automate ETL for open-source data and company fundamentals as a Bloomberg alternative
- Integrating a RAG-based AI framework to enable natural language querying and stock price prediction over large-scale public financial datasets

Stock Prediction Program | Python, JavaScript, HTML/CSS, PyTorch, Flask

August 2025 – October 2025

- Built a full-stack ML application using Flask, HTML, and CSS
- Implemented a multilayer perceptron model with PyTorch and engineered endpoints for data handling, preprocessing, prediction logic, and user I/O
- Designed a multi-page interface for stock input and model inference

TECHNICAL SKILLS

Languages: Java, Python

Frameworks and Tools: Git, PyTorch, AWS, NumPy, pandas, scikit-learn, Flask, React, HTML/CSS

Software Engineering:

- Object-oriented design, unit testing, debugging, performance profiling
- Modular and scalable system design
- Linux command-line, Git version control
- Basic understanding of networking (sockets, routing concepts, APIs)